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QoS-aware Network Slicing and Resource Management for Internet of Vehicles in 5G networks

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ABSTRACT

The development of Internet of Vehicles (IoV) technologies brings with it numerous challenges, such as heterogeneous Quality of Service (QoS) in 5G and beyond (B5G). These challenges go hand in hand with spectrum scarcity, one of the biggest challenges of future wireless technologies. This will be exacerbated especially in the era of Vehicle-to-Everything (V2X) communication due to the limitation of radio resources such as channels, bandwidth, and power. This challenge is further intensified by the increasing number of vehicles. Therefore, spectrum sharing initiatives have a significant impact on traffic safety and efficiency. In response to these challenges and to meet the diverse QoS requirements of vehicular applications within 5G/B5G, new paradigms such as New Radio Vehicle-to-Everything (NR-V2X) and Network Slicing (NS) have emerged as important solutions. Network Slicing efficiently divides the physical network into slices tailored to Ultra-Reliable Low Latency Communications (URLLC), Enhanced Mobile Broadband (eMBB) and massive Machine Type Communications (mMTC). In this paper, we investigate the IoV slicing problem with QoS support, focusing on partitioning the physical network into three different slices: URLLC, eMBB and mMTC. To ensure seamless communication in vehicular networks, our mixed method approach effectively incorporates handover mechanisms, emergency traffic prioritization, based services, and road network-specific parameters. In addition, we propose two resource allocation algorithms that enable efficient allocation of resources to vehicles and comply with the standardization principles of the 3rd Generation Partnership Project (3GPP). These algorithms aim to prioritize emergency traffic during incidents while maintaining an acceptable QoS for non-safety services in a resource-constrained environment, thus improving Key Performance Indicators (KPIs). Detailed simulations show the effectiveness of the proposed algorithms, and confirm their ability to improve the performance of emergency services in terms of end-to-end delay while ensuring acceptable reliability and throughput. Comparative evaluations further highlight the superiority of the proposed NS method over other existing approaches.

1. Introduction

The automotive industry plays a central role in the field of connected mobility in smart cities to improve road safety, operational efficiency and user comfort. In this environment, Vehicle-to-Everything (V2X) communication has become a fundamental communication paradigm within future wireless networks. Its importance lies in providing seamless connectivity and communication between vehicles and the surrounding infrastructure. In particular, V2X communication has emerged as one of the key verticals in the context of Fifth Generation (5G) networks [1]. This underlines its central role in improving the capabilities of intelligent transportation systems and connected mobility solutions. The term V2X covers a range of communication modalities that enable the exchange of information via Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), Vehicle-to-Vehicle (V2V) and Vehicle-to-Network (V2N) communication. V2V communication in particular enables vehicles to exchange important data such as speed, position and status so that they can make better decisions on the road, especially in emergency scenarios [2]. Vehicular communication covers various use cases, which can be broadly divided into safety services and non-safety services (infotainment services). Safety services are at the forefront of V2X communication. They are intended to increase road safety by enabling the immediate exchange of information and messages between road users and the infrastructure. On the other hand, infotainment services provide value-added experiences for vehicle users, such as online gaming, video streaming, and content sharing, all of which contribute to the overall comfort of passengers [3].

Despite the convenience of a unified radio system, the diverse service requirements of IoV applications, which include throughput,

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Table 1
Use cases and service requirements adapted from ETSI/3GPP.

Slice/Service type	Max-latency (ms)	Reliability (%)	Throughput type
URLLC	1–20 ms	90–99, 9%	Low/med/high
eMBB	Up to 100 ms	Fair	Medium
mMTC	1–50 ms	Low to heigh	Low

latency, reliability, and availability, pose significant challenges to resource allocation in vehicular communications. In addition, the exponential growth in the number of connected vehicles is exacerbating the strain on the Vehicular Radio Access Network (RAN) [4]. Conventional RAN resource management approaches are therefore inadequate given the limited bandwidth, diverse services and increasing number of connected vehicles. Against this background, the development of an innovative NR-V2X slicing method is essential. This method must be able to dynamically allocate radio resources to vehicles, offer customized services and efficiently meet various Quality of Service (QoS) requirements. One of the overarching goals of the Network Slicing (NS) design is to enable different virtual networks on a shared infrastructure and to ensure efficient resource allocation while meeting Service Level Agreement (SLA).

The International Telecommunication Union (ITU) has identified three primary use cases for 5G services, each serving specific purposes and applications: Enhanced Mobile Broadband (eMBB), massive Machine Type Communications (mMTC), and Ultra-Reliable Low Latency Communications (URLLC) [5]. The QoS characteristics of the three standardized 5G service types relevant to V2X applications are summarized in Table 1. There is a consensus among telecommunications organizations, including the Third Generation Partnership Project (3GPP) and the Next Generation Mobile Network Alliance (NGMA), on the convergence of these services into a shared physical infrastructure. This means moving away from the traditional approach of using different network solutions for each one. In this context, release 16 [6], introduced NR-V2X and physical layer enhancements to meet stringent 5G QoS requirements and ensure interoperability with various Radio Access Technologies (RATs) such as Long-Term Evolution -V2X (LTE-V2X), Intelligent Transport Systems operating in the 5 GHz band (ITS-G5), and Dedicated Short Range Communications (DSRC). These standards have contributed significantly to the definition of message formats, the standardization of specifications and the definition of requirements for the Intelligent Transportation System (ITS).

With the introduction of release 17 [7], 3GPP has incorporated technical advances to support URLLC applications for V2X services, which are essential for safety-critical functions such as improved position accuracy, collision avoidance, and resource management. The New Radio (NR) side link enhancements facilitate direct communication between vehicles and nearby devices and extend V2X capabilities to autonomous vehicles, remote driving, drone support and holographic services. The implementation of dynamic spectrum sharing in Long-Term Evolution (LTE) and NR optimizes the use of radio resources, reduces interference and improves system performance. Looking ahead to release 18 [8] and release 19 [9], 5G-Advanced evolution within 3GPP includes AI/ML-based services and acts as a bridge toward 6G mobile communications. These releases introduce new features in the areas of Multiple Input, Multiple Output (MIMO), Unmanned Aerial Vehicles (UAVs), extended reality (XR), mobility management, and duplex operation. NS is a potential solution that addresses the above challenges [10] by creating different, logically isolated network resources and functionalities that are tailored to specific requirements and built on a common, programmable physical infrastructure. Furthermore, the integration of 5G technology into the V2X communication framework undeniably represents a significant enhancement to the system, offering increased bandwidth, lower latency, and extended coverage [11].

To this end, our research addresses the development of an adaptive resource allocation strategy that aims to maintain QoS in the face of dynamic and disruptive vehicular accident scenarios. By utilizing NS, physical resources are flexibly allocated to each logical network slice to meet user demands and support different QoS requirements. To address the heterogeneous QoS requirements of various applications in vehicular networks, we consider three distinct types of NS — eMBB, mMTC, and URLLC — to provide customized services, focusing on low latency and high reliability. In addition, we propose and implement two algorithms based on semi-static and dynamic NS methods to efficiently utilize network resources in dynamic contexts while meeting these QoS requirements.

Motivated by these challenges and technological developments, this paper proposes a novel resource allocation framework for vehicular networks that builds upon 5G NR-based NS. Specifically, we design and evaluate two slicing strategies — a semi-static and a dynamic algorithm — to support the heterogeneous QoS requirements of URLLC, eMBB, and mMTC services in an accident-driven vehicular scenario. Our approach integrates multiple system-level considerations such as CQI/MCS, safety prioritization, slice isolation, service type differentiation, and handover management following 3GPP standardization. The main contributions of this paper can be summarized as follows:

- We design a realistic NR-V2X vehicular network model incorporating an accident scenario and evaluate it under dynamic mobility and service heterogeneity.
- We propose a QoS-aware NS and resource allocation framework that accounts for safety prioritization, 3 slices type (URLLC, eMBB, mMTC) based on 3GPP-defined service categories, and dynamic vehicular conditions.
- To ensure efficient resource allocation and high performance of different critical applications in highly dynamic vehicular environments, we develop and implement two algorithms — a semi-static and a dynamic slicing strategy.

The remainder of this article is organized as follows. Section 2 introduces related work and the concepts used in this paper. Section 3 describes the functional behavior of our system the proposed algorithms. In Section 4, we present the simulation framework employed to analyze various application-level QoS metrics for V2X communications, and we detail our proposed model along with the implementation metrics. In Section 5, we present the simulation results and evaluate the reliability, latency, and throughput. In Section 6, we draw some conclusions.

2. Related works

Various studies have investigated NS and resource allocation in vehicular networks from different perspectives. This section provides an overview of different related works. To improve clarity and coherence, we categorize the relevant literature into four thematic areas: (1) QoS management in 5G vehicular networks; (2) Safety-critical services and message dissemination; (3) Network Slicing for resource allocation in V2X; and (4) Algorithmic optimization Approaches for NS. We also highlight how our proposed framework addresses limitations in existing works.

2.1. QoS management in 5G vehicular networks

In 5G NR-V2X, ensuring QoS across slices with heterogeneous service demands is a major challenge. This complexity arises because, in 5G networks, each data packet passes through a specific Packet Data

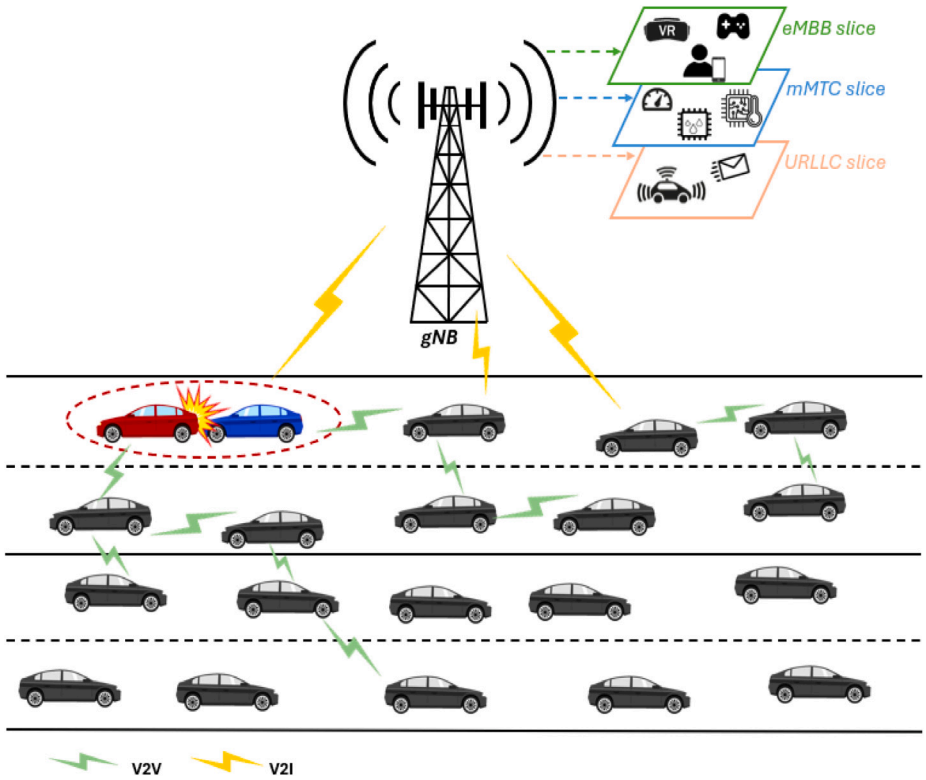

Fig. 1. Illustration of the considered scenario and V2X communication model.

Table 2
The standardized 5QI to QoS features (adapted from [13]).

5QI value	Resource type	Default priority level	Packet Delay Budget (PDB)	Application
1	Guaranteed Bitrate (GBR)	20	100 ms	Conversational voice
2	GBR	40	150 ms	Video live streaming
9	GBR	90	300 ms	Web traffic
86	GBR	18	5 ms	V2X messages

Unit (PDU) and a Data Radio Bearer. Within these pipelines, there may be one or more data flows with different priority levels, data rates, latencies, etc., which are referred to as QoS flows. QoS is a set of technologies and mechanisms that allow network operators to manage and prioritize the delivery of data traffic, ensuring different levels of service for different applications and user experiences [12].

A crucial parameter of this profile is the 5G QoS Identifier (5QI), which serves as a reference for the standardized QoS attributes listed in Table 2. To meet the requirements of the selected 5QI, the network must configure all elements from the wireless physical resources to all physical resources at the Core Network (CN) interfaces. Higher values of 5QI generally indicate more demanding or stricter QoS requirements for a particular data flow.

In order to meet the diverse requirements of vehicles in NR-V2X networks, several studies have been carried out to improve QoS, especially for safety-critical services. For example, Deinlein et al. in [14] presented a scheme to simulate V2I and V2N use cases using the 5G data plane as mobile communication technology. The simulation was implemented using Objective Modular Network Testbed in C++ (OMNeT++) and an extended version of SimuLTE and Veins frameworks. The evaluation showed that V2X packets had higher delays than other applications, despite having the highest priority. This work has evolved into [15] by considering the Medium Access Control (MAC) layer schedulers in the 5G base stations to perform the analysis and obtain the result of end-to-end delay of packets and reliability of data transmission. The researcher considered First-In-First-Out (FIFO), the default priority from the QoS profile, the remaining delay budget, the Channel Quality Index (CQI) status, the number of Hybrid Automatic Repeat Request (HARQ) retransmissions, and scheduling based on packet size. The analysis focused on the average number of vehicles connected to the 5G base station, packet delay in data transmission and reliability, using a motorway simulation. Trivisonno et al. [16] focused on reducing delays and session drop rate in V2X communications. In particular, they focused on the Multi-Level QoS (MLQ) feature to improve the QoS of safety-critical V2X services. A recent study [17] investigated a three-tier vehicular edge-core architecture for vehicles to optimize resource placement. This architecture includes small base stations/access points, macro base stations and CNs, with a focus on QoS-driven management. The proposed approach uses recurrent neural networks to predict resource availability and QoS output. Experiments were conducted in both physical and representative V2X environments, with service requests set to high thresholds to simulate V2X congestion. The use of RDQ3N (integration of RNN and DQN) resulted in remarkable improvements in reliability metrics, latency and resource efficiency, especially for safety-critical intelligent transportation systems and infotainment services.

2.2. Safety-critical services and message dissemination

By leveraging application-driven insights, NR-V2X's capabilities demonstrate its potential to revolutionize emergency services and improve the safety and connectivity of transportation systems. In particular, this enables a range of applications to optimize emergency response, including the warning of hazardous areas and the efficient dissemination of safety messages [18]. In this context, a framework

for the dissemination of safety messages in V2X communication was presented in [19]. Different priority levels were assigned to different applications to reduce delays and increase the reliability of message distribution. Chen et al. [20] presented a Two-Level Adaptive Resource Allocation (TARA) mechanism for secure message transmission in vehicular networks. The TARA mechanism includes resource reservation and allocation modules. The resource reservation module uses a sequential resource allocation model that employs supervised learning for the transmission of different messages. While TARA achieves high packet delivery and lower latency, it sacrifices some flexibility in data transmission. Furthermore, in [21], a road state based on an adaptive NS framework tailored to vehicular networks was presented. This framework focuses on prioritizing emergency traffic in incident scenarios while ensuring acceptable QoS for non-emergency services. The research utilized a Mininet WIFI emulator to build a simulated vehicular network, used the ONOS SDN controller to manage network resources, and implemented the proposed scheme using the OpenVirteX NS tool developed by the Open Network Foundation (ONF) in collaboration with Korea University, Seoul, Republic of Korea.

2.3. Network slicing for resource allocation in V2X

In the resource admission approach, the network infrastructure must assess whether the reservation of a new network slice meets the SLAs of both the new and existing slices. Numerous studies have recognized NS as a critical enabler for resource allocation in vehicular networks [22,23], where 5G NS facilitates V2X services by dividing resources among vehicular devices, the RAN, and the CN. Each slice acts as an independent end-to-end network, tailored to meet the specific service requirements of various in-vehicle applications [24]. In this context, Delgado et al. [25] developed a 5G traffic simulation system to generate traffic data slices with real-world characteristics, particularly on CN. In addition to collecting standard KPI data, the simulator was also able to report the CPU, RAM and storage consumption of each Virtual Network Function (VNF). Caballero et al. [26] addressed RAN slicing for multiple operators with different shares in the network infrastructure and proposed a dynamic mechanism for resource allocation to achieve fairness between network slices. Similarly, Tang et al. [27] proposed an adaptive algorithm for resource allocation strategy in 5G NS, which utilizes a Markov decision process (MDP) with a slice-based virtual resource scheduling scheme and NOMA technology to improve the system's QoS.

Moreover, Khan et al. [28] proposed a model that implements NS for V2X services in the downlink direction. Dedicated network slices are assigned for autonomous driving and infotainment, which are responsible for safety messages and video streaming respectively. The vehicles are grouped into clusters, with a slice leader designated for each cluster to coordinate access. To ensure fair and efficient resource distribution across slices, the authors employed a priority-based slicing management mechanism using Proportional Fair (PF) scheduling. Using the Long-Term Evolution Advanced System-Level Simulator, the proposed framework demonstrated significant improvements in QoS metrics, including reliability and throughput, in a highway scenario with network clusters and dynamic slicing.

Cui et al. [29] introduced the k-means++ method for clustering services based on SLAs, allocated multiple services to each network slice, and proposed the Shared Proportional Fairness Scheme (SPFS) for scheduling slices to improve the fairness of resource utilization. In addition, Cui et al. [30] proposed a long short-term memory-based deep deterministic policy gradient algorithm (LSTM-DDPG) to ensure stable performance for slices and fulfill SLAs. They validated the effectiveness of LSTM-DDPG compared to other machine learning methods in resource allocation.

In [31], the authors proposed an NS scheme tailored to LTE-V2X RSUs for vehicular users, consisting of Local Resource Allocation (LRA) and Shared Resource Allocation (SRA) layers, to optimize the performance of modern vehicular services. The performance evaluation conducted in a controlled laboratory environment showed the superiority of their method over existing algorithms in terms of throughput, end-to-end delay, jitter, and packet loss ratio.

2.4. Algorithmic optimization approaches for NS

Some studies have focused on the algorithmic and mathematical modeling aspects of NS. Various methods have been applied in these approaches, such as linear programming [32], (meta-)heuristics [33] and game theory [34]. A comprehensive overview of various algorithms and strategies can be found in the work by Ojijo et al. [35]

In this regard, Debbabi et al. [36] investigated the architecture of NS, focusing on algorithmic approaches in the context of optimization and operations research methods. Similarly, Chen et al. [37] addressed the optimization of long-term time-averaged total system capacity while meeting the requirements of high reliability and low latency for vehicular communication links. They presented an online algorithm to optimize Resource Block (RB) allocation and power control in each time slot, which contributes to efficient network management in vehicular communication.

In addition, Ge et al. [38] proposed a new NS algorithm tailored to autonomous vehicular networks, specifically addressing the URLLC scheme to improve reliability and delay performance in LTE networks. For the same goal, Shi et al. [39] proposed a solution to the problem of motion planning at intersections using V2V communication by implementing a heuristic motion planning algorithm and utilizing the 5G URLLC slice.

2.5. Summary of related works and research gap

The reviewed literature spans four main thematic domains: (i) QoS management in 5G vehicular networks, (ii) safety-critical services and message dissemination, (iii) NS for V2X resource allocation, and (iv) algorithmic optimization approaches for NS. While each of these studies contributes significantly within its scope, the majority are constrained to LTE-based deployments, do not address emergency traffic, and do not integrate key technical challenges such as handover support, HARQ retransmissions, CQI/5QI-based scheduling, and slice-level queue-aware prioritization.

In contrast, our work presents a unified framework that combines semi-static and dynamic slicing algorithms to manage URLLC, eMBB, and mMTC slices in a differentiated yet interdependent manner. It supports end-to-end slice isolation, integrates HARQ-aware and CQI-driven scheduling strategies, and incorporates handover-sensitive and queuing mechanisms to ensure robust QoS under high-mobility and congestion conditions. The proposed framework is fully implemented using a full-stack simulation environment — OMNeT++, Simu5G, INET, SUMO, and Veins — with algorithmic logic powered by Python modules and aligned with 3GPP-compliant design principles. A comparative analysis of the reviewed works is provided in Table 3, highlighting the gaps addressed by our framework.

3. Proposed network slicing architecture and resource allocation strategy

In this section we describe the problem context and the associated performance requirements, followed by a detailed explanation of the proposed network model. Furthermore, we present and analyze two NS algorithms designed to optimize resource allocation and improve performance in vehicular communication networks.

Table 3
Comparative summary of related works and proposed framework.

Work	Considered key challenges								Contributions	Used methods
	HARQ	HO	SI	Q	CQI & 5QI	ET	SP	RAT		
[14]	✓	✗	✗	✗	✓	✗	✓	LTE-5G	Evaluating the performance of parallel running applications in 5G V2I/V2N traffic scenarios	• Simulation-based evaluation • Extending SimuLTE • Proportional Fair Queuing
[15]	✓	✗	✗	✓	✓	✗	✓	5G	A path forward scheduler performance optimization	• Simulation-based evaluation
[16]	✗	✓	✗	✗	✓	✗	✓	LTE	Leverage 5GS capabilities to interact with Application Functions and adopt the Level of Automation (LoA) concept	• MLQ,LoA concept • On-field measurements • System-level simulation
[17]	✗	✓	✓	✓	✗	✗	✓	5G	Introducing an intelligent agent and slice controller named RDQ3N for safety-critical and infotainment applications	• RDQ3N, LoA concept • On-field measurements • System-level simulations
[19]	✗	✗	✗	✓	✓	✓	✓	LTE	Resource allocation algorithms for LTE-V2X networks to prioritize safety messages	• Simulation-based evaluation
[20]	✗	✗	✗	✗	✗	✓	✓	LTE	Adaptive resource allocation frameworks to support diverse types of safety messages	• TARA framework • A trace-driven simulation
[21]	✗	✗	✗	✓	✗	✓	✓	LTE-A	A fog-level adaptive slicing framework based on SDN/NFV that implements a resource allocation scheme	• Mininet-Wifi, OpenVirteX • ONOS controller, Docker
[26]	✗	✓	✗	✗	✗	✗	✗	LTE-A	Dynamic resource allocation among tenants based on a weighted proportionally fair objective	• Pareto-optimality and Distributed semi-online algorithm • Simulation
[27]	✗	✗	✗	✓	✗	✗	✗	5G	A slice-based virtual resources scheduling scheme with NOMA technology	• MDP and approximate dynamic programming • Simulation
[28]	✓	✗	✓	✓	✓	✗	✓	LTE-A	A network clustering and slicing algorithm that includes the allocation of slice leaders to ensure high-quality V2V and V2I communication links	• PF • Simulation
[29]	✗	✗	✓	✗	✗	✗	✓	5G	A slice coordinator with clustering, scheduling, and resource allocation for optimized weight distribution and load tolerance	• K-means++ and SPFS • Simulation
[30]	✗	✗	✓	✗	✓	✗	✗	5G	A QoS-guaranteed NS orchestration	• LSTM • DDPG
[31]	✗	✗	✗	✗	✓	✗	✗	LTE	Dynamic resource allocation method from a Virtual Resource Pool (VRP)	• Simulations and real-world test
Our Work	✓	✓	✓	✓	✓	✓	✓	5G	Semi-static and dynamic slicing algorithms that ensure QoS-aware resource allocation for three slice types: URLLC, eMBB, and mMTC in realistic V2X scenarios involving congestion, mobility, and emergency message handling.	• Simulation based: OMNeT++, SUMO, Simu5G, INET, Veins • Python-based control logic • 3GPP-compliant simulation

SI = Slice Isolation, Q = Queueing, ET = Emergency Traffic, SP = Safety Prioritization, RAT = Radio Access Technology.

3.1. Problem description and performance requirements

V2X technology supports a wide range of applications, including safety-related services, traffic management, and infotainment. As vehicular networks evolve, the introduction of 5G technology with its transformative capabilities, namely URLLC, eMBB, and mMTC, plays a critical role in supporting these services. A representative scenario that illustrates the practical importance of these capabilities arises in traffic accidents, where fast and reliable communication is essential.

The convergence of requirements for low latency, massive connectivity, and high data rates underlines the need for an adaptable and

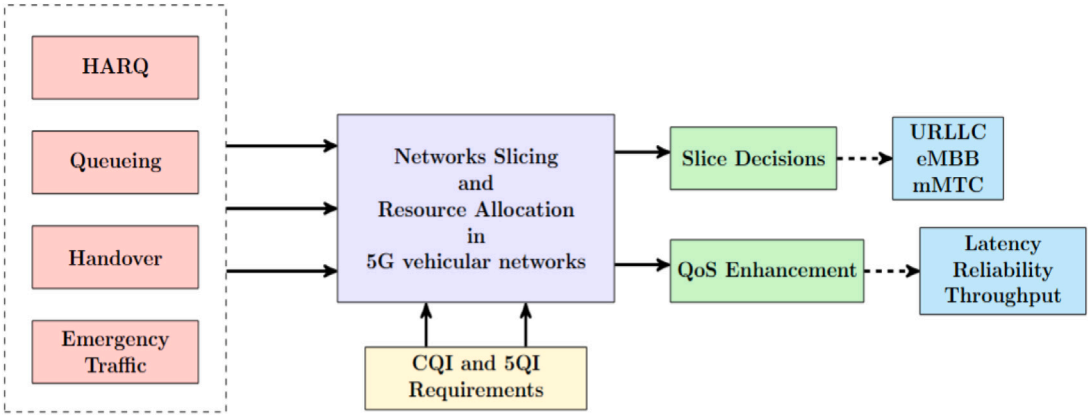

Fig. 2. Overview of the proposed model.

robust network infrastructure. To address these challenges, our simulation framework is designed with two primary objectives: integrating NS and improving QoS. To ensure differentiated service delivery, we consider three standardized slice types:

- Slice 1 (URLLC): Reduces end-to-end latency and increases the robustness of transmission participants. It plays an important role in the safety of vehicles.
- Slice 2 (eMBB): Enables access to bandwidth-intensive content and ensures the seamless downloading of high-quality movies within seconds.
- Slice 3 (mMTC): Enables seamless communication between a variety of devices and promotes the efficient exchange of information.

In order to efficiently provide these customized services with varying QoS requirements, SLAs are associated with each slice to guarantee specific performance levels. Fig. 1 provides a visual overview of the considered vehicular scenario and communication model. As shown, both V2I and V2V links coexist. A collision event (accident) is highlighted as an example use case where URLLC traffic is prioritized. The figure also illustrates how the three slice types are logically separated to support distinct applications, ranging from safety services (URLLC) to entertainment (eMBB) and telemetry (mMTC), all managed via gNB infrastructure.

3.2. Network model

Considering the research question and the direct relationship between resource allocation, NS, and QoS, we propose a method to address the increasing demand for bandwidth resources in 5G-based vehicular environments. This challenge is of particular importance due to the growing demand for reliable and efficient communication. Key issues such as dynamic resource allocation, seamless handover, efficient resource sharing, infrastructure utilization, and spectrum allocation significantly complicate the interaction between users and networks.

To address these challenges, we propose a QoS-aware NS scheduling scheme based on NR-V2X, which introduces slice-specific resource handling and queuing mechanisms while adapting to vehicular mobility. Inspired by recent literature and aligned with 3GPP standardization, our scheme integrates additional parameters, such as HARQ processes, MCS levels, and CQI-based prioritization, to support diverse application requirements across URLLC, eMBB, and mMTC slices. These elements are schematized in Fig. 2, which illustrates the operational workflow of the proposed framework. As shown, inputs such as HARQ operations, queuing decisions, mobility-induced handovers, and emergency traffic are fed into the central resource management module. This module performs NS and resource allocation in 5G vehicular networks. The outputs of this module include slice decisions targeting URLLC, eMBB, and mMTC services, as well as QoS enhancement actions aimed at maintaining service-level guarantees in terms of latency, reliability, and throughput. This architecture highlights the integration of QoS-driven metrics and traffic awareness to enable intelligent, slice-specific resource control dedicated to an accident scenario as an emergency case.

In order to meet the different QoS requirements for each network slice, we have implemented individual queuing systems that enable slice-specific traffic isolation, prioritization, and resource management. This approach improves system latency-aware behavior and ensures compliance with service-level guarantees. As illustrated in Fig. 3, individual buffers are assigned to each slice (Slice 1, Slice 2, and Slice 3), allowing traffic to be classified and temporarily stored independently before processing. These buffers are related to an admission control and slice allocation module, which evaluates the resource availability, current buffer occupancy, and slice priority levels to determine whether packets are accepted for scheduling. By isolating traffic at the buffer level and applying slice-specific allocation policies, this mechanism ensures that latency-sensitive applications receive timely access to transmission resources while maintaining fairness across slices.

Since different slices require different network resources, the implementation of queues proves helpful in organizing and allocating resources based on the specific requirements of each slice. In addition, the use of queuing systems enables isolation between slices. This is a crucial element in avoiding interference and ensuring that the characteristics of one slice do not adversely affect the performance of the others. For example, in a scenario such as an accident, Ultra-Reliable Low Latency Communications traffic related to safety messages may require immediate processing, while Enhanced Mobile Broadband traffic for infotainment purposes may tolerate some delays.

Secondly, we have integrated a QoS module that can include parameters such as 5QI and QoS QFI to facilitate the management and differentiation of QoS for each slice type (URLLC, eMBB and mMTC). URLLC applications are labeled with the highest 5QIs. In addition, enabling the QoS flows with the 5QI values listed in Table 2 can ensure that the KPIs for different application types are met.

When allocating resources with NS, we have two options: static and dynamic. With static slicing, each Virtual Network (VN) receives a fixed portion of physical resources that remains constant over time, providing a predictable and stable environment for vehicular services [40]. This is important for applications that require consistent and reliable connectivity, such as autonomous driving and real-time traffic management. This simplifies configuration for network administrators and ensures clear separation between services. However, fixed spectrum allocations can lead to under- or over-utilization of resources. In addition, it does not match the changing resource requirements of services over time, resulting in over-provisioning of resources. To address these limitations, we have introduced a semi-static slicing method that provides a degree

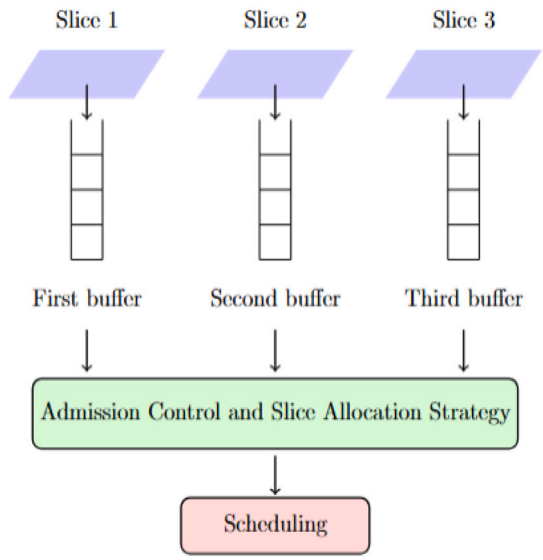

Fig. 3. Admission control mechanism.

of flexibility to better manage resource allocation. In this context, we propose a comparative analysis of semi-static and dynamic slicing methods to evaluate their effectiveness.

Moreover, we consider the effects of handover, the process of disconnecting from the current base station and establishing a new connection to one with higher signal strength. Handover plays a crucial role in enabling seamless transitions between base stations and ensures an uninterrupted connection as vehicles move through the network. This mechanism contributes to reliable and continuous communication for connected vehicles and improves the overall performance of the vehicular network.

Finally, on the receiver side, it is crucial that the packets are received correctly. The hybrid automatic repeat request, a combination of Forward Error Correction (FEC) and Automatic Repeat Request (ARQ), increases reliability. Confirmation takes the form of an ACKnowledgement (ACK) or a Negative ACKnowledgement (NACK), which confirms receipt or error. In the subsequent Transmission Time Intervals (TTIs), the Media Access Control scheduler prioritizes the HARQ buffer checks and allocates the resources accordingly. Some modifications allow 16 simultaneous packet reception processes per HARQ buffer and reduce the HARQ ACK/NACK arrival acceptance time to 1 ms.

3.3. Network slicing algorithms

With the rise of artificial intelligence (AI), particularly machine learning and deep learning, researchers are applying these techniques to improve results on difficult tasks where traditional human-developed algorithms fail. Despite the potential benefits of 5G service assurance, the lack of a public dataset of real 5G network quality assurance data poses a challenge. Collecting such data for label training is time-consuming and costly. To address these issues, we opt for algorithms with proactive components as a simpler and more feasible approach.

3.3.1. Semi-static slicing algorithm

We assume that N_{gNB} represents the number of gNodeB. The total available bandwidth is given as BW MHz and the quantity V = 1,2,...,100 represents the number of vehicles. The algorithm includes several calculations, such as the determination of the channel width *(chWidthA)* and the modulation and coding scheme *(mcsA)*. The algorithm includes parameters that are defined at the beginning and also named in Table 5. To distinguish the symbols on the receiver side, a normal cyclic prefix (cpA) is added to each OFDM symbol, which is a copy of the last part of the OFDM symbol.

Therefore, the minimum channel width *(chWidthA)* is determined by capturing the interplay between the above key parameters in determining the minimum viable channel bandwidth in vehicular networks based on 5G technology. The channel bandwidth (BW) is determined by the following formula:

$$chWidthA = N_{\mathrm{RB}} \times N_{\mathrm{SC}} + R \times SCS + Gb \times 2 \tag{1}$$

where:

N_{RB} is the Maximum Transmission Bandwidth according to [41]

N_{SC} is the Number of Subcarriers per resource block

SCS is the Subcarrier Spacing

Gb is Minimum guard band for each UE channel bandwidth

In this semi-static solution, the user's bandwidth requirement is met by a proportional, fair distribution of resources among users, with the available RBs distributed fairly among all UEs, even when different applications are active. Furthermore, we have considered handover to increase the utilization of radio resources by selecting the UEs undergoing handover according to the Received Signal Strength Indicator (RSSI) based on the algorithm proposed in [42]. The data rate is determined using the equation given by 3GPP in [43].

The complete procedure of this method is presented in Algorithm 1: Semi-Static Network Slicing and Resource Allocation (SSNSA), which is a low-complexity, slice-based resource allocation procedure. It starts by taking the total available bandwidth (BW), the number of gNBs (base stations), incoming resource block requests ($RB_{request}$), channel quality indicators (CQI), and 5G NR parameters such as subcarrier spacing (ScsA) and guard band (Gb) as inputs. In each iteration involving vehicular users, when an RB request arrives, the algorithm first checks for handover conditions (RSSI threshold) and performs a handover if necessary to maintain connectivity. If no handover is required, SSNA computes the necessary channel bandwidth and selects the MCS based on the reported CQI. The algorithm then applies a priority-based slicing management strategy with a Proportional Fair (PF) resource allocation approach: it reserves a fixed URLLC bandwidth slice of the total spectrum and distributes resources (RBs) to vehicular users according to that slice. Finally, SSNA calculates the key performance metrics, end-to-end latency (D) and reliability (Rel), from the allocated resources and returns the channel bandwidth.

3.3.2. Dynamic slicing algorithm

The dynamic slicing algorithm is initiated by initializing some parameters and configuring the guard interval (giA), which is set to the minimum value denoted by 'g' to maximize the spectral efficiency according to 3GPP 38.101.1–1 Table 5.3.3-1 [41]. After initialization of the parameters and handover verification, the maximum MCS is calculated with CQI. The algorithm is then called to determine the minimum usable channel bandwidth with the function chWidthA(dataRateA, mcsAmax, g).

The algorithm uses the network conditions and takes into account the achieved performance, such as SLA compliance and the Packet Error Probability (PEP) at regular intervals, denoted by T. PEP serves as a critical measure of the reliability and performance of the communication system. The Pe_threshold, which represents the threshold value for the acceptable packet error probability, is determined based on the predefined QoS requirements and performance targets. In this case, a Pe_threshold of 0.01 (or 1%) has been set to ensure the robustness of the system by allowing only a minimum error rate, preventing possible corruption or loss of data. The probability of packet errors that occurred in each data flow during the last interval T is calculated and compared to a threshold value:

$$P_e = \frac{TxPackets - rxPackets}{txPackets} \leq 0.01 \tag{2}$$

The defined behavior is represented by the state machine in Fig. 4. For each T, the needed bandwidth is recalculated and multiplied by

Algorithm 1: Semi-Static Network Slicing Algorithm

Input: BW, N_{gNB}; $RB_{request}$; CQI; cpA; ScsA; Gb; txPowerA
Output: chWidthA; mcsA; D; Rel; Throughput

```
Initialization:
N_gNB ← 8
ScsA ← 15 kHz ·2^μ    /* μ ∈ {0,1,2,3} */
Gb ← 805 kHz
for all v = 1 to V do
    if receive RB_request then
        /* Checking for handover conditions */
        if Handover_Conditions(RSSI_threshold) then
            Perform_Handover()
        end
        else
            chWidthA ← Calculate_Channel_Bandwidth(dataRate, N_RB, N_SC, SCS, Gb)
            mcsA ← Calculate_MCS(CQI)    /* Calculate MCS using CQI */
            Resource_Allocation_Strategy ← Proportional_Allocation    /* Initialize Resource Allocation Strategy */
            URLLC_Bandwidth_Slices ← Allocate_Bandwidth_Slice(BW, chWidthA, Resource_Allocation_Strategy)    /*Allocate URLLC
              Bandwidth Slices */
            Allocate_Resources_For_Vehicular_Users(URLLC_Bandwidth_Slices)    /* Allocate resources for vehicular users based on
              URLLC_Bandwidth_Slices */
        end
    end
    Calculate D, Rel    /*Calculate Latency and Reliability using (5,6)*/
    return chWidthA
end
```

Table 4
Comparison of semi-static and dynamic slicing strategies.

Evaluation metrics	Semi-static slicing	Dynamic slicing
Resource allocation strategy	Fixed per-slice resource reservation based on predefined priorities	Adaptive reallocation of resources based on network conditions
Adaptation to channel conditions	Limited	High
Suitability for URLLC	High under average conditions; may underperform during surges	High (low-latency prioritization)
Suitability for eMBB	High (stable allocation)	High
Suitability for mMTC	Sufficient	Efficient
Scalability	High (due to simplicity)	Moderate to high (depending on update frequency)
Best use scenario	Moderate-load, predictable environments	High mobility, high density and emergency scenarios

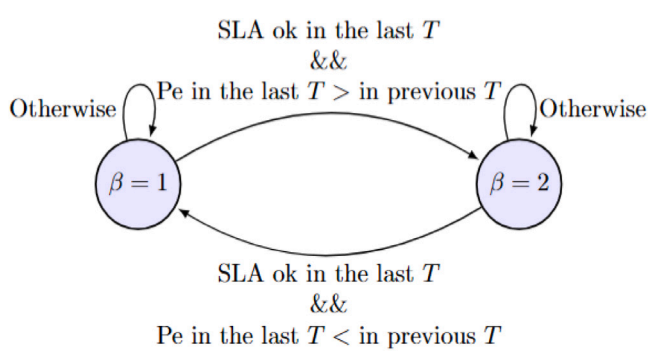

Fig. 4. State machine implemented by the dynamic algorithm for resource allocation Slice.

the value β which can be 1 or 2, depending on whether the bandwidth needs to be duplicated or not. Once the bandwidth is determined, the MCS is determined.

At the same time, three virtual buffers are set up as queues to process incoming packets and ensure isolation systematically. Packets are only removed during scheduling or handover procedures.

The iterative process within the algorithm comprises several important steps: calculation of the maximum achievable MCS (mcsAmax) based on the CQI, determination of the minimum realizable channel bandwidth (chWidthA), calculation of the minimum MCS (mcsAmin) required to support the desired data rate with the selected channel width and guard interval, evaluate the SLA compliance and PEP changes over the interval T, adjust the bandwidth multiplier (β), recalculate the MCS and channel bandwidth based on the updated parameters, adjust the channel bandwidth and channel number and re-evaluate the MCS to ensure that the required throughput is supported.

The overall goal of the algorithm is to optimize resource allocation by striking a balance between latency and reliability while dynamically adapting to evolving network conditions. Priority is given to selecting the minimum MCS that can fulfill the requirements, preferring the highest feasible MCS based on the CQI. The use of the QoS-aware scheduling metric enables the optimal allocation of RBs to user services, thereby improving the QoS requirements of each service. This approach focuses on minimizing latency, which is especially important for the URLLC slice, while balancing it with periodic HARQ ACKs to mitigate a possible increase in error probability. Algorithm 2 describes the Dynamic Network Slicing Algorithm (DNSA) process for slice A, the URLLC slice.

The DNSA adaptively allocates wireless resources based on real-time network feedback and vehicular conditions. For each vehicle, the algorithm begins by checking for any pending RB requests. If such a request exists, it first verifies whether a handover is necessary based on

received signal strength (RSSI) and executes the handover procedure if the condition is met. Once stable connectivity is ensured, the algorithm computes the maximum supported MCS from the CQI and determines the minimum required channel bandwidth to support the vehicle's data rate. It then selects an appropriate MCS value within the feasible range. During the periodic control loop, DNSA uses SLA performance indicators, specifically compliance and packet error trends from the previous interval, as described previously. After these adjustments, DNSA schedules the appropriate slice-level resources, assigns them to the vehicle, and removes the transmitted packet from the corresponding queue. Finally, latency and reliability metrics are computed for performance monitoring.

3.4. Expected performance of semi-static and dynamic slicing schemes

The proposed semi-static and dynamic network slicing algorithms are designed to optimize resource allocation under different operational conditions. The semi-static strategy is tailored for moderately varying vehicular traffic scenarios, where stable communication and low computational overhead are prioritized. This approach leverages pre-configured CQI/MCS thresholds and greedy handover handling to ensure basic QoS guarantees with minimal resource reconfiguration.

On the other hand, the dynamic slicing algorithm is developed to address environments characterized by rapid traffic fluctuations, dense vehicle presence, and critical QoS requirements such as those seen in mission-critical and emergency-driven vehicular applications. This approach dynamically adapts slice configurations, enabling fine-grained control over latency, reliability, and bandwidth allocation.

The expected operational characteristics and trade-offs of both approaches are summarized in Table 4. This comparison provides a hypothesis-driven basis for performance evaluation, which is further analyzed in Section 5.

Based on these comparative characteristics, we hypothesize that the DNA algorithm will demonstrate superior performance in latency and reliability sensitive services, particularly in URLLC and time-critical eMBB scenarios, due to its real-time adaptation capabilities. Conversely, the semi-static approach is expected to remain effective in relatively stable or low-mobility environments where computational simplicity and basic QoS guarantees are prioritized. These expectations will be empirically validated through the performance evaluation presented in Section 5.

4. Simulation and implementation

This section presents the simulation tools, system modeling strategy, and details of the implementation used in our study.

4.1. Simulation tools

There are a limited number of 5G simulators that model the 5G NR system in detail based on the current 3GPP standards and work at packet level. Notable simulators include 5GLENA [44] based on ns3 [45,46], 5GK-Simulator2 [47], Vienna 5G SL simulator [48], WiSE [49], 5G Toolbox by Matlab [50], and Simu5G [51]. This paper focuses on Simu5G, which is based on the SimuLTE library [52], a well-established framework in industry and academia. Simu5G is based on the OMNeT++ [53] simulation framework, which is one of the top options for public network simulators, and provides a set of models with defined interfaces to create complex simulation scenarios. It utilizes OMNeT++, a versatile discrete-event simulation framework applicable to various networks. Simu5G, which integrates the models [54], simulates 5G RAN and CN and supports Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD) modes with different Next Generation Node Bs (gNBs).

In addition, Simu5G can be easily integrated with the OMNeT++ vehicular mobility libraries such as Veins [56], Simulation of Urban Mobility (SUMO) [57] and Artery [58]. It includes realistic channel models

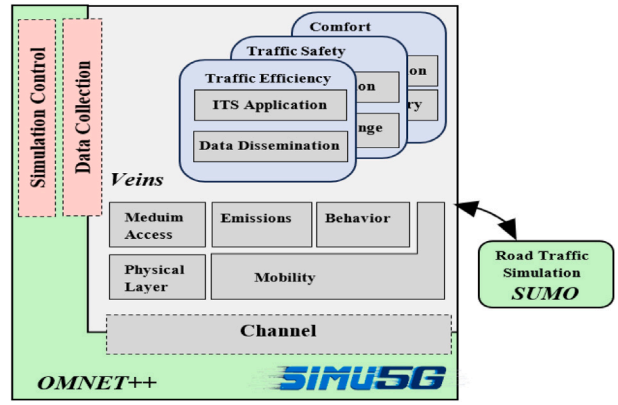

Fig. 5. Framework Architecture (adapted from [55]).

and provides resource scheduling for uplink and downlink, including carrier aggregation and multiple numerologies. Simu5G supports diverse User Equipment (UE) mobility models, including vehicular mobility, and can operate in real-time emulation mode, enabling interaction with real devices through OMNeT++ real-time event scheduling and INET library configuration for IP packet exchange.

In this simulation study, we used a combination of powerful tools to carefully model and analyze a 5G vehicular network, focusing on an accident scenario. The simulation framework was mainly implemented using Omnet++, coupled with Veins to exploit the communication capabilities between vehicle and infrastructure and among vehicles, as well as SUMO and SIMU5G, as shown in Fig. 5. Furthermore, the current version of Simu5G does not include NS, so the NS algorithms were implemented in Python in addition to the functions already available. The visualization of the simulation results was performed with the popular data plotting library Matplotlib [59]. The synergy of Omnet++, Python and Matplotlib enabled the creation and investigation of three different network slices.

4.2. Modeling

Addressing the URLLC latency challenge requires a strategic modification to the basic components of 5G. In particular, the scheduling of radio resources needs to be recalibrated to align with the demanding latency specifications. In this work, we have focused more on the RAN and considered vehicular networks for the provision of different services. Our simulation was developed for the performance evaluation of QoS requirements in the context of vehicular communication based on data exchange over the 5G-Uu interface and/or the PC5 interface.

As shown in Fig. 1, we consider the IoV to consist of base stations and vehicles. We have set up a Vehicular Ad Hoc Networking with 100 vehicles (V), 8 gNBs, 3 servers, 1 router and 1 User Plane Function (UPF) for the data plane. In addition to the RAN, the UPF plays a crucial role in the management of user data and PDU sessions. The UPF also plays an important role at user plane level. It is responsible for managing user data and PDU sessions. It is also responsible for routing and forwarding traffic, including traffic control and redirection as well as QoS handling (enforcement of bit rates, packet marking, packet-to-QoS flow mapping). In order to provide different services to the vehicles, the shared network infrastructure is divided into several logically isolated networks, such as slice 1, slice 2, and slice 3 using Python. For simplicity, we assume that a vehicle has only one service and can only access one slice. In addition, the local resource blocks are managed by the resource layer.

To simulate an accident scenario within this vehicular model, we implement a forced deceleration and stoppage of one or more vehicles in SUMO at predefined locations, emulating a collision event. This stop event is synchronized with OMNeT++ via the TraCI interface and interpreted as an accident occurrence by the network layer. Upon

Algorithm 2: Dynamic Network Slicing Algorithm

```
Input: Initialize all the parameters
Output: chWidthA; mcsA; D; Rel; Throughput
Initialization:
Gb ← g
β ← 1
for all v = 1 to V do
    if receive RB_request then
        if Handover_Conditions(RSSI.threshold) then
            Perform_Handover()
        end
        else
            mcs_max ← Calculate_MCS(CQI)    /* Calculate MCS using CQI */
            chWidthA ← Calculate_Channel_Bandwidth(dataRate, N_RB, N_SC, SCS, g, mcs_max);
            mcs_min ← MCSmin(dataRateA, chWidthA, g)    /*Determine the minimum MCS*/
            if mcs_max > mcs_min then
                mcsA ← mcs_min + 1    /*Set initial MCS index*/
            end
            else
                mcsA ← mcs_max
            end
        end
    end
end
LOOP Process:
if SLA OK in last T && Pe in last T > in prev. T then
    β ← 2    /*Increase bandwidth if SLA not met and Pe increased*/
end
if SLA OK in last T && Pe in last T < in prev. T then
    β ← 1    /*Maintain bandwidth if SLA met and Pe decreased*/
end
mcs_max ← MCSmax(rxPowerAmin, vehicularParams)    /*Recompute max achievable MCS considering vehicular-specific parameters*/
chWidthA ← CBwmin(dataRateA, mcsAmax, g)    /*Recompute minimum viable channel bandwidth*/
chWidthA ← chWidthA ×β    /*Adjust bandwidth based on SLA*/
mcs_min ← MCSmin(dataRateA, chWidthA, g)    /*Recompute minimum MCS*/
if mcs_max > mcs_min then
    mcsA ← mcs_min + 1    /*Adjust MCS if necessary*/
end
else
    mcsA ← mcs_max
end
Allocate_Resource_To_Vehicle(sliceA, vehicleID)    /*Allocate resources to the vehicle from slice A*/
Remove_Packet_From_Queue(slice_queue, packetID)    /*Remove packet from the queue*/
Calculate D, Rel    /*Calculate Latency and Reliability using (Eq 5,6)*/
```

receiving this mobility state change, the affected vehicle initiates the generation of URLLC traffic representing a safety-critical alert. These messages receive the highest priority and are placed into the URLLC queue, where the Simu5G slice manager schedules them preferentially. In accordance with established V2X safety communication practices, we assume that vehicle-mounted communication units remain operational in the moments following impact, thus enabling the timely transmission of critical data. This integration ensures a more realistic simulation of accident-induced URLLC communication within our 5G NR-V2X framework.

The simulated scenario represents a linear highway segment where 100 vehicles are inserted at fixed entry points and follow predefined routes. Traffic is generated through flow-based insertion, and vehicle mobility is governed by SUMO's default microscopic car-following model to ensure realistic highway dynamics. Two main scenarios were considered during the evaluation process: a baseline scenario without incidents and an accident scenario in which a vehicle collision is explicitly triggered using the TraCI interface by forcing an abrupt vehicle stop. However, only the accident scenario is presented and analyzed in this article, as our primary objective is to evaluate QoS performance under emergency conditions where reliable and low-latency communication is critical. Each simulation is executed for 600 s and repeated 10 times using different pseudorandom seeds. Performance metrics — specifically throughput and end-to-end latency — are measured for each of the three 5G network slices (URLLC, eMBB, and mMTC) and averaged across the ten independent runs for analysis.

All described components of our simulation are implemented and integrated in OMNET++ to monitor and record the Key Performance Indicators for each slice.

4.3. Implementation

OMNeT++ provides support for various transport layer protocols, including Transmission Control Protocol (TCP), User Datagram Protocol (UDP) and Stream Control Transmission Protocol (SCTP). These protocols serve as basic elements for various applications and services in vehicular networks [60]. When choosing the transport protocol, we opted for UDP. UDP offers minimal overhead and is therefore suitable

for low-latency applications. However, it does not always provide the required level of reliability, which is solved by using ARQ mechanisms.

In addition, the spectrum allocation standardized by 3GPP for communication in 5G-based vehicular networks covers a wide range of frequencies:

- Frequency Range 1 (FR1): This range extends from 410 MHz to 7.125 GHz and includes frequencies traditionally used for cellular communications, such as sub- 6 GHz bands like 600 MHz, 700 MHz, 800 MHz, 900 MHz, 1.8 GHz, 2.1 GHz, 2.3 GHz and 3.5 GHz [61].
- Frequency Range 2 (FR2): This range covers the millimeter wave bands extending from 24.25 GHz to 52.6 GHz, providing high bandwidth for specific 5G applications [61].

Our simulation uses the 2.1 GHz and 5.9 GHz frequency bands and uses FDD for V2I and TDD for V2V communication to optimize resource allocation. FDD, with its wide coverage and reliable penetration, is ideal for V2I scenarios and ensures robust connectivity for eMBB and mMTC slices in long-haul communications. Conversely, TDD enables dynamic time slot adaptation in the 5.9 GHz band and is therefore well suited for the intermittent and low-latency requirements of V2V URLLC traffic.

On the other hand, the required channel bandwidth depends on various factors, such as the Modulation and Coding Scheme (MCS), the channel conditions and the QoS requirements. The choice of MCS depends on the channel quality. The CQI value guides the selection of the MCS to optimize the data transmission efficiency. It is determined by sensing the channel feedback and is based on the Signal-to-Interference-plus-Noise Ratio (SINR) [62], which is calculated as follows:

$$SINR = \frac{P_{RX}^{BS}}{\sum_i P_{RX}^{i} + R} \tag{3}$$

where:

P_{RX}^{BS} is the power received from the serving GNB,

P_{RX}^{i} is the power received from the ith interfering GNB,

R is the Gaussian noise

P_{RX} is calculated as follow: $$P_{RX} = P_{TX} - P_{loss} - F - S \tag{4}$$

where:

P_{TX} is the transmission power,

P_{loss} is the path loss,

F, S are the attenuation caused by fading (fast and slow fading)

The CQI is used to determine the modulation scheme (Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (16 QAM), 64QAM, 256 QAM) and the code rate using tables from [43]. These values can be used to determine the MCS index, which is defined by Tables 5.1.3.1–1 and 5.1.3.1–2 in [43].

As for Subcarrier Spacing (SCS), which refers to the spacing between subcarriers in the frequency domain, its value determines the bandwidth of the subcarriers. It is a key factor in defining the numerology of the system, which in turn affects the symbol duration, subframe duration and other parameters in the time domain. NR-V2X supports multiple SCSs such as $15\,\text{kHz} \cdot 2^{\mu}$, where μ is the SCS configuration and is given by $\mu = 0, 1, 2, 3$ [63]. The configured simulation parameters are summarized in Table 5.

5. Performance analysis

In this section, we discuss the simulation results and make a qualitative evaluation considering latency and reliability. To evaluate the performance of our simulation, we determined QoS parameters such as reliability, throughput, and latency for each slice type under semi-static and dynamic NS methods for spectrum allocation.

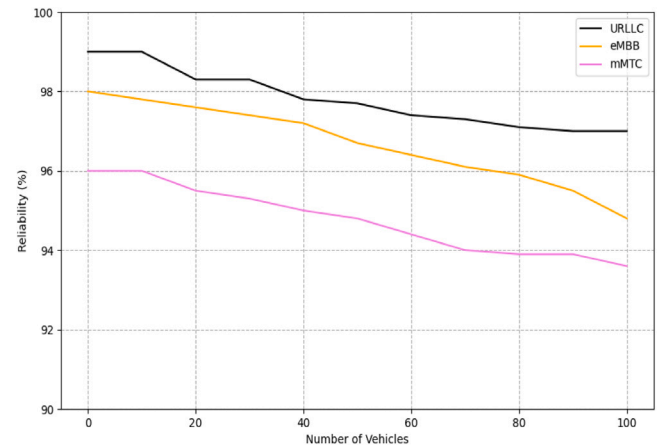

Fig. 6. Reliability variation versus number of vehicles under DNSA.

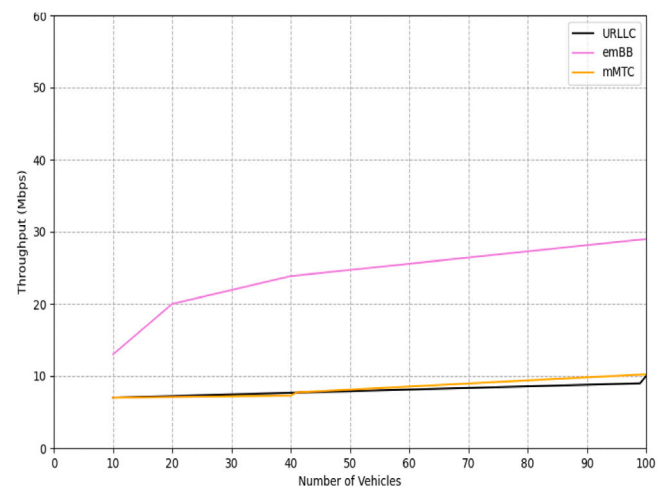

Fig. 7. Throughput variation versus number of vehicles under DNSA.

Table 5
Simulation parameters.

Parameter	Value
Simulation time	600 s
Number of servers	3
Number of gNBs	8
gNB height	25 m
gNB transmit power	46 dBm
Vehicle antenna height	1.5 m
Number of vehicles	100
Vehicles transmit power	26 dBm
Average speed	80–100 km/h
Carrier frequency	2.1 GHz 5.9 GHz
Duplex mode	FDD TDD
Transmission Time Interval (TTI)	1 ms
Subcarrier Spacing (SCS)	30 kHz
Bandwidth	50 MHz 30 MHz
MCS	According to [43]

5.1. Reliability analysis

The first analysis aimed to evaluate the reliability, a percentage value that indicates the success rate of packet transmission within the PDB. When calculating reliability within the simulation model, lost packets are also taken into account, which increases the significance of this metric. After the arrival of each packet and the calculation of the latency using Eq. (5), it is determined which PDB is adhered to. The calculation is based on the following formula:

$$Rel(PDB, App) = 1 - \frac{OutPDB(PDB, App) + LP(App)}{LP(App) + totalRecPack(App)} \tag{5}$$

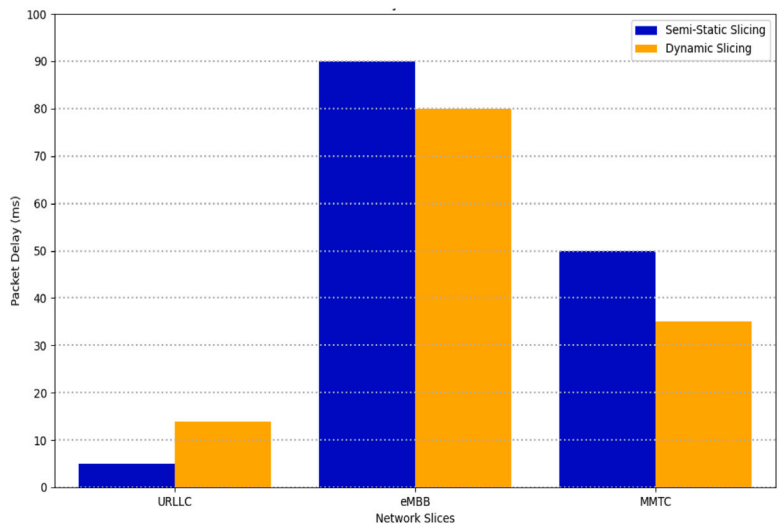

Fig. 8. Latency variation.

where:

PDB is the latency value in seconds for which reliability calculation is performed,

App is the corresponding application,

$OutPDB$ is the number of packets that did not meet the maximum latency value PDB,

LP is the total number of lost packets per application

$totalRecPack$ is the total number of received packets per App

Fig. 6 shows the simulation results of the reliability variation under DNSA. The reliability for each slice was computed as the percentage of packets meeting the PDB, averaged over 10 simulation runs. This algorithm adaptively allocates resources among the URLLC, eMBB, and mMTC slices based on traffic load and service-specific QoS requirements. The results indicate that the network generally provides satisfactory performance, varying reliability across slices as vehicle density increases.

In particular, URLLC shows the highest reliability values, which are close to 99%. Even with an increasing number of vehicles, URLLC communication maintains a relatively high reliability with a mean of approximately 98.0% ±0.3%. This performance is attributed to the prioritized resource allocation and reduced congestion in URLLC-enabled slices. However, as vehicle density increases, the reliability of the URLLC begins to decline, reaching 97%±0.3% at 100 vehicles. This degradation is attributed to increasing congestion, which leads to interference, latency, and packet loss, thereby compromising the reliability of safety-critical communication.

eMBB communication has also consistently high reliability (between 98%±0.7% and 95.5%±0.7%) due to robust network coverage and minimal congestion. Nevertheless, its reliability is slightly lower than that of URLLC, which is due to the less stringent reliability requirements of eMBB. As the number of vehicles increases, the reliability of eMBB gradually decreases. Unlike URLLC, eMBB applications can tolerate a certain level of packet loss or latency, thereby mitigating the effects of lower reliability.

In contrast, mMTC reliability is lower compared to URLLC and eMBB (between 96%±0.9% and 93.6%±0.9%), as resources are dynamically redirected toward more delay-sensitive slices. Nonetheless, mMTC reliability remains relatively stable.

5.2. Throughput analysis

Another challenge addressed by dynamic slicing is maintaining consistent throughput across heterogeneous service types.

As shown in Fig. 7, which represents the Throughput variation versus the number of vehicles under DNSA algorithm. Under the highest load scenario, the URLLC slice achieved a mean throughput of 10±0.2 Mbps. The throughput of mMTC also follows a similar fluctuation and generally increases as the number of nodes increases, reaching 9±0.3 Mbps. The eMBB slice achieved a higher average throughput of 28±0.7 Mbps.

The observed improvement in throughput with an increasing number of nodes indicates efficient resource management that meets communication requirements between different slices. In particular, the throughput requirements for mMTC and URLLC are lower than for eMBB applications, since URLLC prioritizes low latency and reliability over pure throughput, while the eMBB slice is designed for high data rates and capacities to enable bandwidth-intensive applications such as high-definition video streaming, virtual reality, and augmented reality. Consequently, the slightly higher deviation observed for eMBB is attributed to the fluctuating data rates typical of broadband applications, but the variation remains within acceptable limits. The results confirm that the proposed QoS-aware slicing mechanism efficiently allocates bandwidth across services while maintaining predictable and scalable performance.

5.3. Latency analysis

The impact of semi-static and dynamic slicing on latency is crucial for evaluating network performance. Therefore, packet latency, a key performance indicator for URLLC, is used to evaluate this impact.

$$Del_i = A_i - T_i \tag{6}$$

where:

Del_i represents the calculated latency value in seconds for the ith packet that arrives at the receiver side ,

A_i is the arrival time of packet i,

T_i is the transmission time of packet i

Fig. 8 shows the simulation results for the variation of latency. Each network slice type (URLLC, eMBB and mMTC) is represented by two lines denoting the semi-static and dynamic variations. These lines illustrate the fluctuation of packet delay within each slice type under

semi-static and dynamic conditions. It is noteworthy that the dynamic slicing of URLLC exhibits higher packet latency (9±2.5 ms) than its semi-static counterpart (5±1 ms). This discrepancy results arise from the pre-allocation and configuration of resources in semi-static slicing, which ensures minimal packet delays. Conversely, dynamic conditions lead to fluctuations in network traffic and resource allocation, leading to increased packet delays for URLLC slices. However, these delays remain relatively low compared to other slice types. Both the semi-static and dynamic fluctuations do not exceed 20±1 ms.

eMBB slices have moderate packet delays under semi-static conditions (90±1.5 ms) as they prioritize high data rates and throughput over ultra-low latency. However, since resources are allocated dynamically, packet delays are more consistent compared to semi-static conditions (80±0.5 ms) and do not exceed 100 ms. This improvement is due to better bandwidth utilization under dynamic conditions.

The mMTC slice also benefits from dynamic slicing, showing lower average latency (35±3 ms) compared to semi-static (45±4.9 ms). Since mMTC is less sensitive to delay, both values remain acceptable as they do not exceed 50 ms.

Overall, the results confirm that dynamic slicing improves latency consistency and responsiveness for eMBB and mMTC, while semi-static slicing remains preferable for achieving minimal latency in ultra-critical URLLC applications.

The results confirm the expected reduction in latency for V2X packets when using the dynamic slice QoS scheduler, which provides greater efficiency and better fairness between slices and does not exceed the standards compared to the values in Table 1.

5.4. Comparison of proposed and reference schemes

For the reference schemes, the proposed Dynamic Network Slicing Algorithm (DNSA) is compared with two policy settings: (1) Reliable Software-Defined RAN (RSDR) approach presented in [64], and (2) our Semi-Static Slicing Algorithm (SSNSA), which applies a priority-based slicing mechanism combined with PF scheduling.

The parameters used for performance evaluation include average delay, throughput and reliability, as shown in Fig. 9 in parts (a), (b) and (c), respectively. As can be seen from the simulation results, the proposed scheme outperforms both the RSDR and SSNSA algorithms. In particular, DNSA achieves an average delay of 14 ms for the URLLC slice, which is 1 ms and 21 ms lower than the average delay of SSNSA and RSDR, respectively. For the eMBB slice, the improvements by our scheme were 10 ms and 84.09 ms better than SSNA and RSDR, respectively. For the mMTC slice, the proposed scheme is more successful than the other two systems by 16.25 ms and 42.5 ms, respectively. Fig. 9 b shows that the DNSA reliability for URLLC, eMBB and mMTC slices is up to 90, while RSDR provides the lowest reliability values. Fig. 9 c illustrates the average throughput, which represents the efficiency of bandwidth resource allocation and scheduling for each slice. The comparison with the reference systems shows that the DNSA improvements are very remarkable with 8.015 Mb/s, 24.28 Mb/s and 8.35 Mb/s for the URLLC, eMBB and mMTC slices, respectively; however, SSNSA and RSDR delivered lower throughput.

In conclusion, the proposed scheme effectively meets the average delay, reliability and throughput requirements in all three slices. In contrast, the other schemes do not fulfill the requirements of the different slices, especially in scenarios with a larger number of vehicles that require more resources from the gNB. If critical information is not prioritized in such scenarios, this can lead to detrimental results.

These results are consistent with the hypotheses discussed in Section 3.4. As expected, the DNSA algorithm shows superior adaptability and performance in scenarios characterized by dynamic traffic patterns and strict QoS requirements, especially for URLLC and delay-sensitive eMBB services. Meanwhile, the SSNSA approach shows reliable behavior under stable conditions with moderate load, where its simplicity and low computational overhead are beneficial. This agreement between expected and observed performance underpins the appropriateness of each method in its respective context and highlights the effectiveness of the proposed framework in addressing heterogeneous vehicular communication requirements.

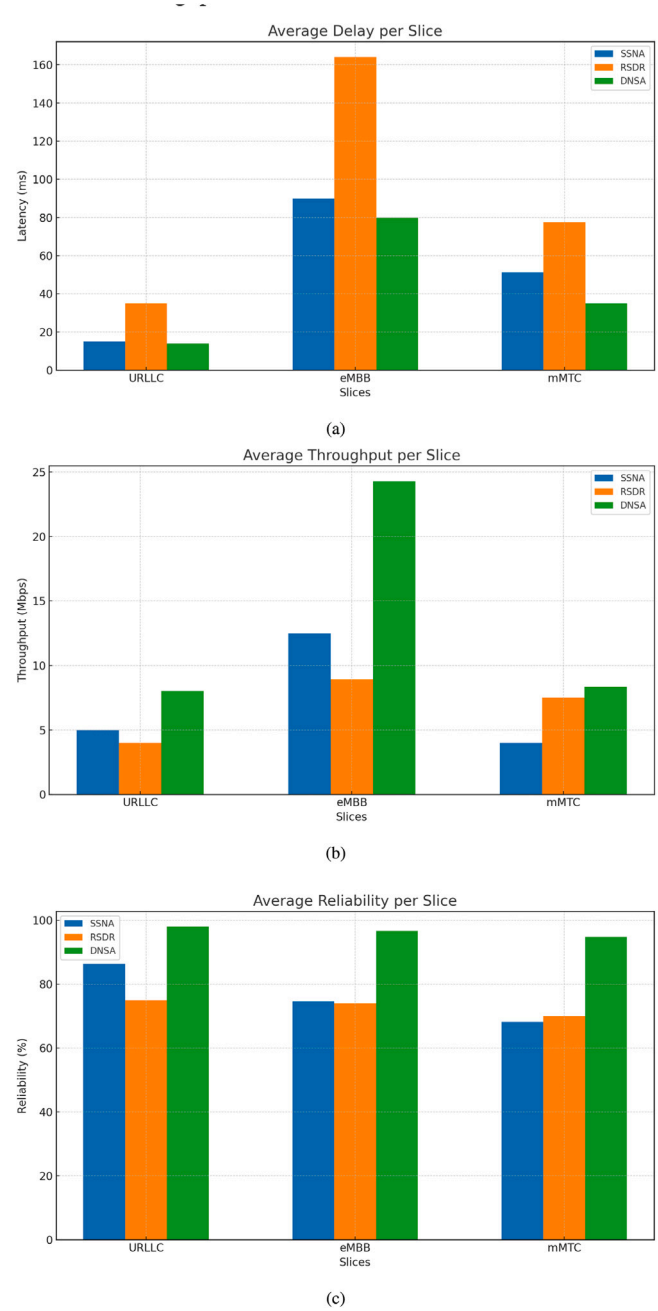

Fig. 9. Performance evaluation results of (a) Delay, (b) Throughput and (c) Reliability between the proposed and reference schemes.

6. Conclusion and future work

In this paper, we investigated a resource management scheme for the V2X slicing environment. Our solution mainly focused on resource allocation with the goal of reducing latency and improving reliability. The problem solution was formulated using two different types of slicing (semi-static and dynamic) for three types of slices (URLLC, eMBB, mMTC). The proposed scheme was coupled with an efficient QoS consideration. Furthermore, a simulation was performed using simu5G, OMNeT++ and related frameworks to ensure that the simulation accurately reflects the impact of the QoS parameters on the

overall performance of the 5G vehicular network. To evaluate the proposed solution, we considered an accident scenario. The simulation results show that NS effectively improves the performance of 5G V2X communication. While the semi-static method provides greater fairness between slices, the dynamic approach has shown greater effectiveness by reducing latency and improving reliability by exploiting variations in wireless channels and network conditions.

Future work will include further analysis considering additional performance metrics and real-world scenarios to provide a more comprehensive understanding of the network's behavior and its suitability for vehicular communication applications. In this context, we could utilize different scheduling techniques and technologies and privacy and security for NS in V2X.

CRediT authorship contribution statement

Wafa Hamdi: Writing – review & editing, Visualization, Software, Formal analysis, Writing – original draft, Validation, Methodology, Conceptualization. **Orhan Dağdeviren:** Validation, Project administration, Writing – review & editing, Supervision, Conceptualization. **Hasan Bulut:** Writing – review & editing, Supervision, Conceptualization, Validation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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